PROJECT AND TEAM INFORMATION

Project Title

Campus Lost and Found

Student/Team information

Team Name	**solvers**
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PROPOSAL DESCRIPTION

Motivation

Every day it's common for students and teachers to lose their valuable belongings on college like mobile phones, wallets, ID cards, laptops, water bottles, books and more. Currently there is no centralized or organized solution for this problem on campus, students resort to the WhatsApp group, help desk or notice boards which are highly inefficient and time consuming.

Our Campus Lost & Found System, named **campus lost and found**, aims to solve this real world campus problem by providing an interactive, dedicated digital platform where students and staff can:
i) Report lost items with detailed description and last known location of the item. ii) Report found items so that the rightful owner can be contacted.
iii) Search and filter through active listings quickly and easily. iv) Track the status of a reported item from active to claimed to resolved. v) View contact information of the reporter to coordinate item recovery.

This project is important because it directly impacts the daily lives of students. Losing important belongings can cause significant distress, a centralized system reduces time and effort-required to recover lost item.

State of the Art / Current solution

Current methods used on campuses

- WhatsApp groups: Students post photos of lost or found items in class or college groups which get no responses quickly.
- Security Desk: Some campuses like ours, Graphic Era University, have a security desk that collects found items, but there is no database or online interface for students to query.
- Word of Mouth: The most common and least reliable method, dependent on social networks.

None of these solutions offer a searchable database, status tracking, or a formal reporting mechanism. **Campus lost and found** fills this gap with a structured, accessible web-based platform.

Project Goals and Milestones

Initial Goal: Build core Lost & Found System

Develop a fully functional web platform where students can report lost and found items. The initial version includes a form to submit item details, a listing page to browse all reports, and a SQLite database to persist data. The FastAPI backend handles all API requests and data storage.

Mid-Term Goal: Add search, filter and status tracking

Enhance the application with advanced filtering by item type(lost/found), category, and status (active, claimed, resolved)

Final Goal: Add image upload and user authentication

Deliver a fully production-ready application by adding two major features. First, allow users to upload images of lost or found items so they can be verified visually. Second, implement a user registration and login system so that only verified campus students can report or manage items, adding accountability and security to platform.

Project Approach

Frontend Development

- HTML5: Page structure, forms, and semantics layout for reporting and listing items
- CSS3: Responsive design using Flexbox and CSS Grid, modal dialogs, animations, and card-based item layouts
- Vanilla JavaScript (ES6+): Fetch API for communicating with the backend, dynamic rendering of item cards, filtering logic, and form handling without page reloads

Backend Development

- FastAPI (Python): RESTful API with endpoints for Create, Read, Update, and Delete (CRUD) operations, Pydantic models handle input validation automatically
- Jinja2: Template engine that allows FastAPI to serve the main HTML page directly
- Uvicorn: ASGI server that runs the FastAPI application locally

Database

- SQLite3: Serverless, file-based relational database. Requires zero configuration and is built into Python. Stores all item records in single lost_found.db file
- Parameterized SQL queries are used throughout to prevent SQL injection attacks

System Architecture (High Level Diagram)

Campus Lost & Found System (KhoyaPaaya)

[USER / BROWSER]

HTML + CSS + JavaScript

Forms | Item Cards | Filters | Modals |

↓ HTTP Requests (JSON) ↑ HTTP Responses (JSON)

[BACKEND SERVER]

FastAPI (Python) + Uvicorn + Jinja2

REST API Endpoints: GET | POST | PUT | DELETE

Pydantic Models for Input Validation

↓ SQL Queries ↑ Query Results

[DATABASE]

SQLite3 (lost_found.db)

Table: items (id, item_name, category, description, location, date_reported, status, reporter_name, reporter_contact, item_type)

Project Outcome / Deliverables

The outcome of this project is a fully functional, browser-based Campus Lost & Found web application. It includes:

- A working web application accessible via any browser at http://localhost:8000

- FastAPI-powered REST API with fully functional endpoints (GET, POST, PUT, DELETE)

- SQLite database with automatic schema initialization and persistent data storage
- Full CRUD functionality: Create, Read, Update, and Delete item reports
- Responsive frontend with item reporting forms, item listing cards, search and filter system

Assumptions

- Users are assumed to be honest i.e. they will report for the lost item whenever they found one
- The system is designed for campus level use with small to medium number of concurrent users.

References

1. FastAPI Official Documentation - Modern, fast web framework for building APIs with Python. URL: https://fastapi.tiangolo.com
2. SQLite Official Documentation - Serverless, self-contained SQL database engine. URL: https://www.sqlite.org/docs.html
3. Uvicorn Documentation - Lightning-fast ASGI server for Python. URL: https://www.uvicorn.org
4. Pydantic Documentation - Data validation and settings management using Python type annotations. URL: https://docs.pydantic.dev
5. MDN Web Docs - Reference for HTML5, CSS3, and JavaScript (Fetch API, DOM manipulation). URL: https://developer.mozilla.org
6. Jinja2 Documentation - Full-featured template engine for Python. URL: https://jinja.palletsprojects.com